Source: Igini M., 2019

From Ornamental Seeds to Fuels:

The Complex Dynamics of Palm Oil in Indonesia's Socioeconomic Landscape

Energy in the Global South – B265609

The Growing Role of Palm Oil in Biofuels

Take a moment to observe your daily life. Your favorite chocolate bar, the ice cream you enjoy while thinking about the weather getting hotter, the instant noodles you eat, the margarine you use to avoid cooking oil, and even the soap, detergent, and cosmetics you use—many of these products likely contain palm oil (Alonso-Fradejas et al., 2017; Borras et al., 2016). Even more, palm oil is increasingly being used for biofuels as feedstock for net zero energy transition (especially sustainable aviation fuel and biodiesel) and co-firing for coal (Zahid et al., 2024). Palm oil, once primarily used for human consumption, is now increasingly employed in industrial production, particularly biofuels, driven largely by European demand (Borras et al., 2010). Understanding this shift and its societal implications can help us grasp the social and economic impact of biofuel progress on the Global South.

Debates and Perspectives on Palm Oil's Impact

The debates surrounding palm oil remain heated even after nearly a century. These debates touch on issues of equality, human rights, environmental impact, socio-economic challenges, and even the geopolitical colonialism intertwined with this cash crop. The stories and narratives often contradict each other in almost every aspect.

Some argue that palm oil cultivation worsens inequality (Orsato, 2013) and is a leading cause of land dispossession and deforestation (Baka J, 2017; Xin et al., 2021). However, several reports argue that replacing palm oil with other vegetable oils, such as sunflower, rapeseed, or soybean, would only increase deforestation (Chiriacò et al., 2024; Gunarso et al., 2013; Ritchie, 2021). Furthermore, experts identify that new biofuel plantations display noticeable similarities to colonial-era plantations established by multinational corporations backed by colonial powers (Beckford, 1972; Little and Watts, 1994). These discourses arise due to the diverse range of actors and stakeholders involved in the entire oil palm value chain.

Various perspectives bring different interpretations of how natural resources become valuable and meaningful. It is a dynamic process of translation, in which a resource might not be considered a resource until we have developed the knowledge or technology to use it. The translation of natural resources knowledge and practices is often influenced by power dynamics. Certain groups may exert greater influence in shaping dominant narratives and controlling resources, potentially marginalizing alternative standpoints and ways of knowing (Di Giuminiani & Haines, 2020). This process of "translation" is not neutral but deeply intertwined with social dominance hierarchies and the concept of spheres of exchange.

Spheres of Exchange in The Palm Oil Industry

The palm oil industry operates within multiple layers of hierarchy, fostering distinct interactions among various actors, including multinational corporations, policymakers, international consumers, smallholder farmers, indigenous communities, local governments, certification bodies, and financial institutions. These complex relationships form what Jaramillo (2018) describes as "spheres of exchange", which can be broadly categorized into two main groups: the global sphere and the local sphere—categories that we will explore further later in the essay.

Jaramillo explained spheres of exchange as coextensive sets of relationships in which the definitions of objects, their values and notions of property can radically differ. These spheres are organized hierarchically so that the circulations at the top carry more value than those at the bottom. Objects located in different spheres of exchange tend to be incommensurable (things are, therefore, not inter-changeable), for their qualities differ so radically that objects have different potentialities.

The concept of spheres of exchange provides a valuable framework for analyzing the complexities of the palm oil industry in Indonesia. To better understand the dynamics of this industry, we will explore its historical roots and current practices.

Historical Roots of Palm Oil in Indonesia

Indonesia is currently the largest producer of palm oil, supplying 60% of the total global production, followed by Malaysia. These neighboring countries have become the world's leading producers and exporters of palm oil, accounting for more than 80% of the global market since the 1980s (Varkkey, 2012). Despite its dominance in Southeast Asia, palm oil is native to West and Central Africa. The spread of oil palm trees to other regions, including Southeast Asia, was facilitated by colonialism (Zuckerman, 2021; Robins, 2021).

Palm oil prospers on high precipitation, sufficient sunlight, and humid climate – this means the best growing areas are around the equator. In the early 1900s, the demand surge could not rely on the West and Central Africa, and this led to the development in Southeast Asia, where the climate is more favorable, added with its abundant cheaper land and labor. The expansion of plantations was often driven by foreign investment and colonial interests. By the 1930s, the Netherlands Indies and British Malaya had become the world's leading palm oil producers (Wahid et al., 2005; PASPI, 2014).

In Indonesia, the Dutch expansion of oil palm was not explicitly framed as an intentional purpose to colonize the territory for resource extraction. The origins traced back to four oil palm seeds brought by Dr. D. T. Pryce, which consists of two Bourbon seeds from Mauritius and two Dura-type seeds from Amsterdam to be used as exhibits at Bogor Botanical Garden in 1848 (Hunger, 1924; Rutgers et al., 1922). These seeds were later distributed across Java, Sulawesi, Borneo, Nusa Tenggara, Maluku, and Sumatra, initially planted as ornamental plants and "test locations,".

However, some scholars suggest that this seemingly benign introduction of oil palm might have had underlying intentions. As Li (2021) argues, plantations are not merely agricultural enterprises but machines for assembling land, labor, and capital under centralized management to maximize profit. They also act as political technologies that order territories and populations, create new subjects, and reshape landscapes. This perspective raises concerns that the Dutch cultivation of oil palm, particularly in regions like Sumatra, was far from accidental and instead reflected broader colonial ambitions to reorganize territories for economic and political control.

Transmigration Schemes and the Nucleus Estate and Smallholders (NES) Program

This demand surges further supported with the initiatives from Indonesia government and the world bank to expand oil palm plantation to less populated area like Sumatra and Borneo, this came as a transmigration scheme. NES (Nucleus Estate and Smallholders) supported by the World Bank with significant funding since 1970s. The program provided transmigrant families with land for growing both tree crops and food crops, as well as village facilities such as schools, health facilities, and housing materials (Weaving, 1993).

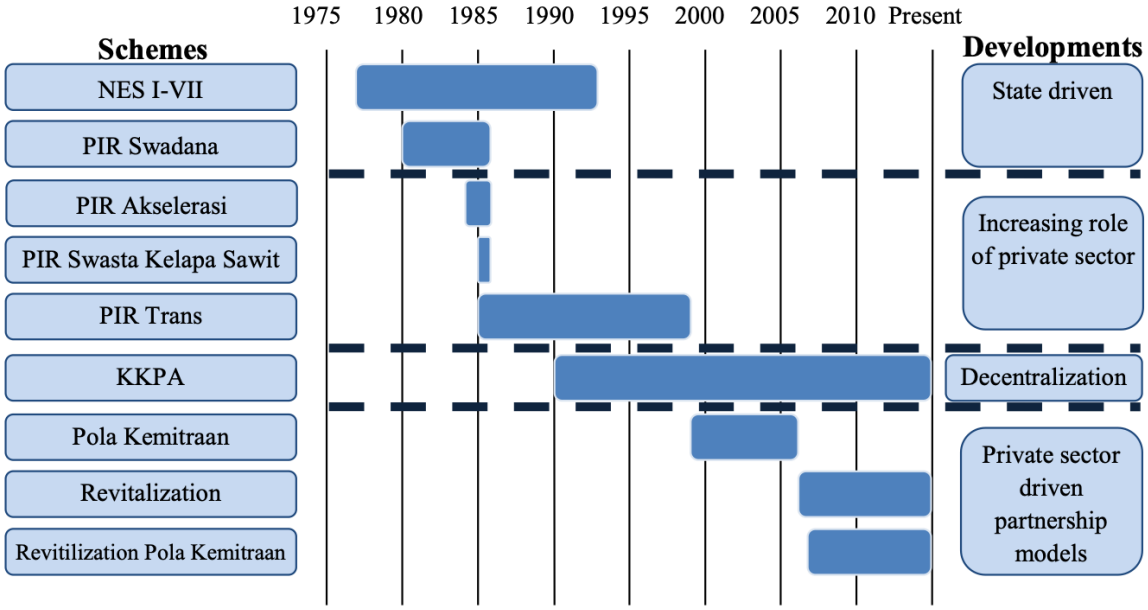
One of the programs called NES III projected that the average net per capita income of smallholders in Danau Salak would increase from approximately $83 to $285 (in constant 1979 valuation) upon full development. At that time, $1 USD was equivalent to Rp. 625 IDR. Adjusted for inflation using the 2024 exchange rate of around Rp. 15,500 per $1 USD (a 25-fold increase), the projected income per capita would be approximately $7,125—significantly higher than Indonesia's current GDP per capita of $4,940 USD based on World Bank data. This figure seemed overly ambitious at the time and gave the impression that the project was overpromising, potentially enticing the government and smallholders to participate in the program with unrealistic expectations.

The Indonesian NES program, designed to boost rural incomes, encountered significant challenges during its implementation. The rapid rollout of multiple complicated projects outpaced the capacity of the public sector estate companies responsible for managing the program. Overstretched and under-resourced, these companies struggled to provide the effective support necessary for smallholder development which critical to the program's success (Weaving, 1993). These challenges significantly hindered the NES program's results in improving rural livelihoods.

Land Conflicts and the Evolution of Plasma Schemes

As more land is contested for further development, land conflicts have become increasingly prevalent. These disputes often arise between government-supported transmigrants and indigenous communities over land rights. Indigenous populations may not possess formal land titles, making their claims more vulnerable in such conflicts. Among the key drivers of these disputes are issues related to the rights to plasma estates. "Plasma schemes are partnerships where smallholder farmers work with large oil palm companies. The companies act as the 'nucleus,' providing support, while the farmers, known as 'plasma,' grow the crops. To date, nine plasma schemes have been developed in Indonesia, as shown in Figure 1 (Firdaus, 2024):

Figure 1. Overview of major Indonesian smallholder schemes



Source: International Finance Corporation, Diagnostic Study on Indonesian Oil Palm Smallholders, 2013

These policy initiatives aimed at expanding palm oil cultivation often framed under the noble objective of improving rural quality of life. The first legal framework for plasma schemes was introduced through Presidential Instruction Number 1 of 1986, which focused on plantation development under the NES-Transmigration Program. This regulation marked the first official effort to govern plasma schemes. Since then, the legal framework surrounding plasma schemes have evolved (Firdaus, 2024).

Biofuels and the Global Political Economy of Palm Oil

From a broader perspective, Indonesia's policy initiatives reflect an alignment with the global political economy trends, where traditional local resources are being turned into global commodities, with biofuels having significant implications. The biofuels revolution emerged as a response to an assumed 'energy crisis' driven by rising fossil fuel costs, particularly in the Global North as they push to reduce their dependence on Middle Eastern oil and ensure 'energy security'. Simultaneously, biofuels represent a new frontier of profitability for agribusiness and energy sectors grappling with productivity challenges and rising costs (Houtart, 2010; McMichael, 2010).

In fact, biofuels are also seen as a solution to environmental concerns, shown as a pathway to transform energy-use trends while maintaining economic growth. This 'win-win' narrative has been a dominant theme in global policy debates, seen in discussions in Europe (Franco et al., 2010) and the United States (Hollander, 2010; Gillon, 2010), shaping the global debates on biofuels (Borras et al., 2010).

Biofuel ambitions of the Global North have created significant pressure on land use in the Global South. Research revealed that, in order to meet Northern fuel needs domestically, it would require converting 70% of European farmland to fuel crops, as well as utilizing the entire US corn and soy harvest, a scenario which seems impossible (Holt-Gimenez, 2007). Consequently, this

demand has shifted the expansion focus to countries like Indonesia, where oil palm plantations rapidly expanded to meet the global demand for biofuels.

Policies such as the EU's biofuel targets, coupled with subsidies for agribusiness corporations, have encouraged large-scale land investments, positioning Indonesia as a key supplier of biodiesel. Together with Malaysia, Indonesia targeted to supply 20% of the EU's biodiesel needs (Holt-Gimenez, 2007). This development has entrenched Indonesia's role in the global biofuel market while raising concerns about the sustainability of such expansion.

While these initiatives offer economic prospects, they raise critical concerns about food security and rural livelihoods. As vast areas of land are converted to oil palm plantations for biofuel production, critics argue that these developments reduce land availability for local communities, undermine subsistence farming, and prioritize global market demands over sustainable local development (Weis, 2010).

Palm oil biofuel's economy, in the Indonesian context, is dominated by multinational corporations, including Cargill, the world's largest private company; the ADM-Kuck-Wilmar alliance, the world's leading biofuels manufacturer; and Synergy Drive (Borras et al., 2010). These corporations exert immense control over production and trade, shaping Indonesia's role in the global biofuel industry.

Global Demand and Its Impact on Indonesia's Role in Biofuel Production

In response to rising biofuel demand, producing countries such as Indonesia have expanded their downstream industries (Singagerda et al, 2018). This has led to a significant increase in palm oil production, from 32.27 million tons in 2005 to 39.42 million tons in 2006, contributing to the broader upward trend from 22.33 million tons in 2000 to 79.07 million tons in 2020 (Ritchie et al., 2023).

To meet the biofuel needs of the Global North, more palm oil is being produced and used for the biofuel, which has pushed food prices higher. Unilever has highlighted the social implications of this shift, noting that palm oil serves as a staple cooking oil for approximately two billion of the world's poorest people. When prices increase, it becomes harder for the lower economy people to manage their living costs, making it worse while they're already limited on resources. These price hikes also trigger production costs for other goods, creating a domino effect that deepens economic inequality (Orsato et al., 2013). This underscores the complex relationship between energy policy, food security, and social equity within the global biofuel discourse.

The palm oil rapid expansion is consequently creating multiple and ambiguous social formations and is affecting different classes in different ways. Independent peasants in the frontier areas where plantations are being introduced might welcome them as a development option or might resist them and defend their land and forests against palm oil corporations. Palm oil smallholders in the plasma scheme have accepted palm oil but have new issues relating to prices, debt and infrastructure. Plantation workers within the palm oil industry have different issues concerning wages and working conditions. The multiple livelihoods related to palm oil expansion thus create the basis for different but interconnected struggles and social movement (Pye, 2010).

Case Study: The NES-Transmigration Program in Mentawai

Taking the example of the NES-Transmigration program in Mentawai, an area in Sumatra, Indonesia. The initiative involved relocating 9,000 transmigrant households and preparing land for plantation development. To facilitate this, indigenous communities were approached and asked to hand over their land to two companies that had already secured Plantation Business Licenses (IUP). These licenses covered areas of 14,157 hectares and 2,542 hectares, designated for the development of transmigration areas and plantation estates (Firdaus, 2024).

Companies approached the indigenous communities with a "sosialisasi" (an act of promotion or dissemination) to persuade them to hand over their lands. Some community members agreed, hoping for improved livelihoods and job opportunities, while others resisted, fearing the loss of their ownership rights. Those who refused argued that, under Indonesia's Basic Agrarian Law, their lands would become state-controlled and granted to companies under HGU (Right to Use) permits (Firdaus, 2024).

The situation got worse for those who handed over their lands, as they were never informed that they would permanently lose ownership. With the lands in the company's hands, they used the land certificates as collateral to secure investment loans for the plantations, leaving the communities without access or control over the properties (Firdaus, 2024). This issue aligns with the concept from Baka (2017) where making space for biofuels is often linked with problematic processes of land grabbing and land dispossession.

Conflicting Dynamics and Local Responses to Palm Oil Cultivation

On the other hand, not all local communities view oil palm cultivation negatively. Some have voluntarily shifted from farming other commodities to start cultivating oil palm after knowing its potential benefits. This transition is often motivated by the higher profitability of oil palm plantation, mainly because of its efficiency, especially labor requirements, compared to other crops like paddy rice (Purba et al., 2019). In addition to its labor efficiency, oil palm is very adaptable to different physical land and climate conditions, enabling it to replace traditional crops such as rice in certain regions and having a more stable yield. Along with its good price stability and financial returns, oil palm is an attractive option for farmers (Edison, 2020).

This shift has impacted the livelihoods of local communities, making them more attached into the oil palm or cash crop economy. Local communities now have increased desires and monetary needs, leading to a greater interest in education, healthcare, and tertiary goods such as motorbikes and electronics (Levang et al., 2016). This is contrasting to what is often narrated by the NGO and public movements that local communities resist the expansion of palm oil plantations and relying on natural resources from forest and subsistence farming (Hasudungan and Neilson, 2020). While in reality, a lot of farmers opt for more profitable crops to enable the modernization of their lifestyle. (FAO, 2016). These changes happen along with the weakening of indigenous authority, ethnic group identity, social structures & relationship over generations in the traditional communities (Koczberski and Curry, 2016).

These illustrate the diverse interactions and conflicting dynamics at the local sphere level, where different actors are impacted by the expansion of palm oil cultivation in many ways. Smallholder farmers are attracted to oil palm cultivation due to the potential of higher incomes and improved access to a better lifestyle. Indigenous communities, whose lands are often violated by palm oil

companies, experience disruption to their traditional practices and land tenure systems. While some people of these communities embrace palm oil for its economic benefits, others resist because of fears of land dispossession and negative impact to their cultural and social structures. These complex dynamics reveal how the local sphere of the palm oil industry is entangled by both opportunities and conflicts.

Local and Global Spheres Interconnectedness

These dynamics are linked to what happens in the larger global sphere. Multinational corporations, international consumers, financial markets, and certification bodies are the actors that shaped the global spheres of palm oil industry. Multinational companies controlling vast areas of land dedicated to palm oil production and trade. The financial sector, such as banks and commodity exchange, supports palm oil expansion and financing. Certification bodies like the Roundtable on Sustainable Palm Oil (RSPO) creating sustainability standards aimed at reducing deforestation, protecting workers' rights, and improving community relations. These global forces influence the practices and outcomes at the local level, creating a complex web of interconnected issues where both positive and negative effects are felt by local farmers, workers, and communities.

The connection between the global and local spheres often creates a sense of disconnection for those at the bottom. While the global sphere holds significant influence through decisions, policies, and rules, these actions sometimes don't align well with the local needs. Global spheres actors rely heavily on local stakeholders to achieve their goals, yet many within the local sphere fail to recognize their own bargaining power. This dynamic displayed the overlooked importance of local spheres in a system where global forces often carry more visible influence and value.

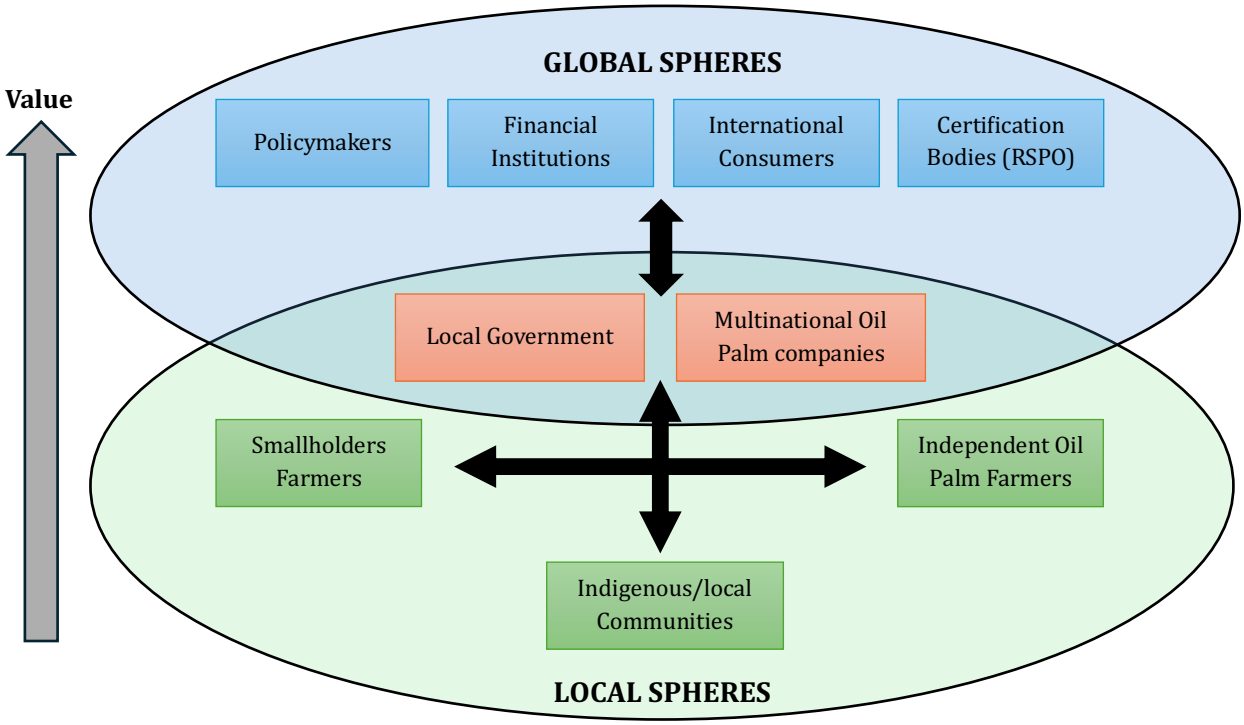


Figure 2. Indonesia's Palm Oil Spheres of Exchange Illustration
Source: Prepared by the author based on Jaramillo (2018)

The Role of Smallholders and the Debate on Monoculture Plantations

The discussion surrounding the palm oil sector often overlooks the social and economic context, in which smallholders and independent oil palm farmers have significant role in shaping international supply chains. This challenges the common perception that Indonesia's palm oil industry is dominated solely by large corporations, with smallholders' involvement seen as involuntary and exploitative. Such perceptions are rooted in historical biases linked to colonial plantations, but they don't accurately reflect the current dynamics on the ground, where smallholder now account for 40.51% of palm oil plantation area in Indonesia (BPS, 2022).

Despite advancement in technology making palm oil production more sophisticated, the role of smallholders and independent farmers remains vital within the social relations of the commodity economic. The debate surrounding large-scale, corporate-run monoculture plantations versus smaller, locally driven alternatives is not as straightforward as it may appear. The reality is often far more complex, involving sophisticated negotiations over issues such as rights, access to resources, labor practices, and livelihoods.

Research by McCarthy (2010) emphasizes that the outcomes aren't just about whether communities are included or excluded. The actual outcomes depend on specific conditions, such as whether there are effective smallholder farming programs, how democratic and fair local village institutions are, where villages and investments are located, and how land ownership and informal land markets operate. These factors determine how communities are integrated into the oil palm economy and whether they benefit from it.

Conclusions

The growing demand for biofuels from palm oil has significantly reshaped Indonesia's socioeconomic landscape, impacting various people in various ways. The historical context reveals its interconnection with colonialism and the global political economy, shaping the Indonesia's role as a key supplier in the international market.

The spheres of exchange concept explain the complex power dynamics and interactions between global and local actors, highlighting the often-unfair distribution of benefits and control over resources. It also uncovers the important involvement of smallholders farmers in the industry, challenging the simplistic narrative of corporate dominance in oil palm cultivation. While biofuels are presented as a sustainable energy future, their expansion raises crucial concerns about socio-economic and environmental changes. Acknowledging these complexities and fostering a more equitable and sustainable approach that prioritizes the well-being of local communities will foster a better future for Indonesia's palm oil biofuels industry.

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